Candidate's examination number……..

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM TWO PRE- MOCK EXAMINATION 2019
CHEMISTRY

CODE: 032 **TIME: 2: 30 HRS**

INSTRUCTIONS

1. This paper consists of section A and B.
2. Answer all questions in both section sections
3. Calculation and cellular / phones are not allowed
4. The following atomic masses may be used :
   C=12, S= 32, N = 14, H = 1, O = 16.

SECTION A: (30%)

1. Write the letter of the correct answer in the box provided for each of the following questions.

(i) A substance that can easily be compressed in its container is:

(A) Greese (B) stone (C) water (D) gas

(ii) Which of the following is not true about hydrogen?
(A)It is lighter than air (B) It burns with blue flame (C) It support combustion (D) it is odourless.

(iii) A branch of science related to the study of matter and its properties is called:-
(A) Physics (B) Chemistry (C) Chemist ((D) Biology

(iv) Which of the following statement is not true
(A) A good fuel should have high energy value
(B) A good fuel should burn with moderate velocity
(C) A good fuel should have an average ignition point.
(D) A good fuel should have a high content of non-combustible materials

(v) A class of fire associated with flammable liquids
(A) Class A (B) Class B (C) Class D (D) Class E

(vi) The process of distillation involves:-
(A) Filtration and decantation (B) Evaporation and filtration

(C)Evaporation and condensation (D) None of the above

(vii) The element which are found in group vii of the periodic table are known as :-
(A)Metals (B)Non-metals (C) Halogens (D)Noble gases

(viii) Bunsen Burner produces a non –luminous flame when
(A) Air holes are closed (B) Air holes are opened
(B) (c) Air holes are half opened (D) It forms soot.

(ix) The maximum member of electrons in the inner most shell of an atom is
(A) 3 (B) 1 (C) 4 (D) 2

(x) The percentage by mass of nitrogen in $(NH_4)_2\ CO_3$ is

(A)28.0 (B)37.5 (C) 29.2 (D) 96.0.

2. Match each item in list A with a response in list B by writing its letter bellow the number of the corresponding item in the table provided.

LIST A	LIST B
(i)Mercury (ii) Group O element. (iii) Luminous flame (iv) Chemical change (v) Identification of Hydrogen (vi)Catalyst (vii) Formed by electron sharing (viii)First Aid (ix) Preventing rusting (x)Solution	A. Galvanization and painting B. Uniform mixture of two or more substances C. Non-metal D. Pop-sound E. Liquid metal F. Produces soot G. New substance is formed H. Covalent bond I. Neon J. A substance that alters the rate of reaction K. A help to a sick or injured person L. Electrovalent bond M. Oxygen N. No new substance is formed O. Water and kerosene P. Periodicity

SECTION B: 70%

3. (a)Define the following terms:
   (i) Chemistry …….
   …….
   (ii) Element …….
   …….
   (iii) Catalyst …….
   …….

   (b)Give the differences between the following: -

   (i) Compound and mixture

Compound	Mixture

   (iv) Suspension and solution

Suspension	Solution

(c)Write the chemical symbols for the following

(i)Tin _____

(ii) Lead _____

4. (a) Define water ……..
……..
(b) Name four different kinds of natural water
   i. ……..
   ii. ……..
   iii. ……..
   iv. ……..
(c)I Is water a compound? Give four reasons.

5. (a)What do you understand by the following terms

(i)Energy ……

(ii)Fuel ……

(iii)Combustion …….

(b)Mention any two characteristics of a good fuel

(i) ……

(ii) ……

(c)For each of the following classes of fire state one burning material

(i)Class A fire ……..

(ii)Class B fire ……..

(iii)Class C fire …….

6. (a)Explain the meaning of the following terms

(i)Water treatment …….
……..
(ii)Water purification …….
……..
(b)State four (4) uses of water in economic activities
   i. …….
   ii. …….
   iii. …….
   iv. …….

(c)Element Q belong to period 3 and group vi of the periodic table.

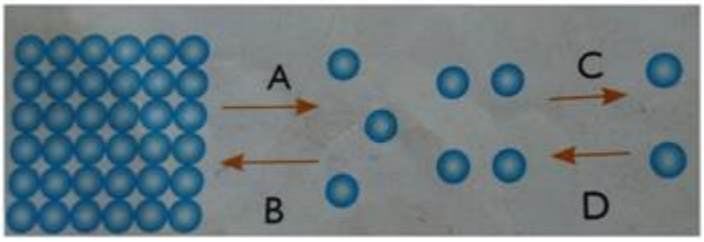
(i)Draw the atomic structure of Q

(ii) Give the atomic number of element Q.

_____

7. (a)Define the following terms:
   (i)Solution _____
   _____
   (ii)Unsaturated solution _____
   _____
   (b)Write the name of the following compounds.
   (i)CUO_____
   (ii)Na_2CO_3_____
   (c)Give out the uses of the following apparatus
   (i)thermometer _____
   (ii)Pipette _____
   (iii)Gas jar _____
   (iv)Tripod stand _____

8. The figure below show the relationship among the three state of matter.



(a)Name the process labelled A to D

(i) A _____

(ii) B _____

(iii)C _____

(iv) D _____

(b)Give two reasons why non luminous flame is more suitable for cooking

(i)_____

(ii)_____

(c)Write down the chemical formula of the following compounds

(i)sodium sulphide_____

_____

(ii)copper (i) oxide _____

_____

9. What do you understand by the following terms?

(i)Empirical formula .....

……

……

(ii)Molecular formula ……..

……

……

(b)A certain compound k contains 15.8% carbon and 84.2% Sulphur. The molar mass of k is 76 g/mol. Determine its

(i)Simplest formula

(ii)Molecular formula

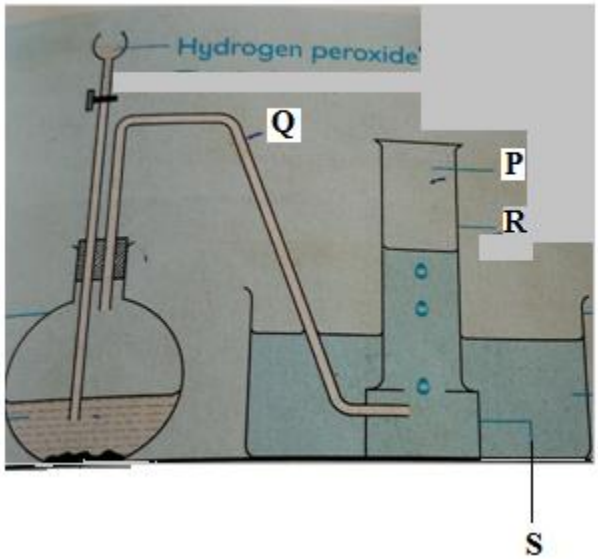
(C)Fill in the missing steps of the scientific produce in the chart below

a

Hypothesis
b
Data colection
Data interpretation
c

(i) a_____
b_____
c_____

10. Study the diagram carefully and then answer the questions that follows



(a)(i) Name the apparatus labelled QR S
Q_____
R_____
S_____

(ii)name the gas P _____
(iii)What is the test of gas P
_____

(b)(i) Mention two properties of gas P _____
(ii)State two uses of gas P _____